

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (EC)(C.S.E.) SEM – II Examination, 2010

Date: 22/06/10

EC - 708 Digital VLSI Design

Time: 1.00 p.m to 4.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

- Q-1 (a) Why Enhancement-Load nMOS Inverter are not used in any large scale digital applications? Draw two different circuit configurations of inverter with Enhancement-type load. 3
- (b) Prove that switching power dissipation of CMOS inverter is proportional to switching frequency. 3
- (c) Discuss the impact of different VLSI design styles upon the design cycle time and the achievable circuit performances. 3
- (d) Implement following Boolean functions. 2
1. $F_1 = (D+E+A)(B+C)$ using full CMOS implementation.
 2. $F_2 = \text{XOR Gate}$ using CMOS transmission gates.
- Q-2 (a) The following parameters are given for an nMOS process: 5
- Substrate doping density $N_A = 10^{16} \text{ cm}^{-3}$, polysilicon gate doping density $N_D = 10^{20} \text{ cm}^{-3}$, gate oxide thickness $t_{ox} = 500 \text{ \AA}$, oxide interface fixed charge density $N_{ox} = 2 \times 10^{10} \text{ cm}^{-3}$. use $\epsilon_{si} = 11.7 \epsilon_0$, $\epsilon_{ox} = 3.97 \epsilon_0$, $kT/q = 0.026 \text{ V}$, energy gap of si = 1.12 eV, Intrinsic carrier concentration $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$, $\epsilon_0 = 8.85 \times 10^{-14} \text{ F/cm}$, $q = 1.6 \times 10^{-19} \text{ C}$. Calculate the threshold voltage V_{T0} .
- (b) Design a resistive-load inverter with $R = 1 \text{ k}\Omega$, such that $V_{OL} = 0.6 \text{ V}$. 7
- The enhancement-type nMOS driver transistor has the following parameters:
- $V_{DD} = 5.0 \text{ V}$
 $V_{T0} = 1.0 \text{ V}$
 $\gamma = 0.2 \text{ V}^{1/2}$
 $\lambda = 0$
 $\mu_n C_{ox} = 22.0 \text{ } \mu\text{A/V}^2$
- (a) Determine the required aspect ratio, W/L
 - (b) Determine V_{IL} and V_{IH} .
 - (c) Determine noise margins NM_L and NM_H .

OR

- Q-2 (a) Explain major types of MOSFET capacitances and show its variations under different operation conditions and physical parameters. 8
- (b) Why is the size of PMOS transistor chosen to be 2.5 times of an NMOS transistor? 4

- Q-3 (a) Consider a CMOS inverter with the following parameters: 8
- nMOS $V_{T0,n} = 0.6V$ $\mu_n C_{ox} = 60 \mu A/V^2$ $(W/L)_n = 8$
- pMOS $V_{T0,p} = -0.7V$ $\mu_p C_{ox} = 25 \mu A/V^2$ $(W/L)_p = 12$
- Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is $V_{DD} = 3.3V$.
- (b) Discuss the different isolation techniques used in fabrication process. 4

OR

- Q-3 (a) Compare the two technology scaling methods. 6
- Constant electric field Scaling
 - Constant Voltage Scaling
- Analytically show how the delay time, power dissipation & power density are affected in terms of scaling factor.
- (b) Explain Interconnect Delay using RC Delay Models and Elmore Delay formula. 6

SECTION-II

- Q-4(a) For a NMOS pass transistor gate terminal is connected to V_{DD} . Give the output voltage range for a square pulse input voltage going from 0 to V_{DD} . 2
- (b) Define the "Euler Path" and significance of it in CMOS technology. 3
- (c) Realize the following Boolean function using TG technology. 3
- $F = AB + AC + ABC$
- (d) Draw the CMOS implementation of the D latch. 3
- Q-5 (a) Explain the 'Domino CMOS logic' with its advantages and disadvantages. How its limitation of cascading stages can be avoided? 9
- (b) Explain Complementary Pass Transistor logic(CPL). 3

OR

- Q-5 (a) Draw Basic BiCMOS circuits and explain static behavior of it in detail. 7
- (b) Draw the equivalent circuits of DRAM and SRAM memory cells. 5
- Compare both type of memory in terms of power consumption and Read-Write speed.
- Q-6 (a) What is the need of Voltage Boot strapping Circuit? Draw Dynamic bootstrapping arrangement and explain it. 7
- (b) Explain static characteristics of the BJT inverter. 5

OR

- Q-6 (a) Write short note on 'first generation' GaAs devices. 8
- (b) Why CMOS technology is not suitable for ultra-fast VLSI circuits? 4